Validation-first measurement of resilience in illicit data economies: a controlled study with a negative-result ledger on public data

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Abstract

Background. Enforcement repeatedly disrupts illicit data-trading ecosystems, yet whether they absorb a shock and resume function is rarely measured under controls able to reject it.
Methods. We state a validation-first protocol: every quantity must survive a control capable of falsifying it. We apply it to the social, financial and infrastructure layers of an illicit data economy, from public archives (the AlphaBay archive, the Elliptic++ Bitcoin graph, the Abuse.ch URLhaus and ThreatFox feeds). Seven controlled analyses share one rule.
Results. Infrastructure is churn-dominant: median online lifetime is **1.36** days, yet **13.88**% of darkened hosts return within one day against a **2.24**% matched floor (**6.2**×). Displacement after the November 2019 Cryptonia–Cannazon shutdown is directional (**+10.5** percentage points on Apollon; **+0.0** on Empire, unmeasurable) but bounded, not estimated. The illicit financial subgraph is peripheral: removing all **14,266** illicit-labelled addresses (**1.734**%) leaves the giant component at **98.88**%.
Conclusions. Of six resilience metrics, none passes all three validity gates. A negative-result ledger records that temporal-graph prediction fails four independent instrument checks through a feature-side separability trap, partially repaired by de-leakage.

Keywords: cybercrime ecosystems; network resilience; measurement validity; darknet markets; temporal graph neural networks; negative results.

1 Introduction

Measuring how a system absorbs a shock and resumes function is hard wherever three conditions hold at once: there is no ground truth, the observations are censored, and the events are too few to estimate from. In such settings a resilience number is only as trustworthy as the control it is run under, and a control that cannot reject anything looks exactly like a control that rejected nothing. We therefore treat that measurement problem — how to construct a falsifiable measurement of resilience in a system with no ground truth, censored data and rare events — as the primary object of study, and apply the resulting protocol to one concrete domain.

Illicit data-trading ecosystems are one such system, and enforcement against them is frequent and visible: in July 2017 Operation Bayonet seized AlphaBay and secretly operated its successor Hansa, and the sites went down. The immediate disruption was total, but what the surrounding ecosystem did next is far less clear, because the archives are right-censored at a single instant, the layers do not share a time axis, and no external ground truth certifies which actors are illicit. We measure all three layers of this economy — social, financial and infrastructure — from public data, and we report the null alongside the estimate wherever a control was able to reject it.

Existing work reports quantities but rarely pairs them with controls shown to be capable of rejecting the naive alternative. Longitudinal measurement established the darknet-marketplace field and its displacement dynamics: the first longitudinal measurement of the anonymous-marketplace ecosystem [1], the collective recovery of dark-web marketplaces after individual closures [2], and vendor migration from AlphaBay and Hansa into Dream Market after Bayonet [3]. The same literature records the absence of any durable reduction in cryptomarket activity after Operation Onymous [4]. It also documents the two mechanisms a measurement has to disentangle: displacement into successor platforms [5], and the reputation-transfer problem that governs whether a vendor can move at all [6].

Table 1: Data sources. Sizes count parsed records in the processed archives. The three layers share no time axis; per-source time coverage, and the Hansa and Valhalla dumps (51,553 reviews and 16,511 listings, from which no reported quantity is drawn), are detailed in Supplementary Note S5.

Layer	Source	Size	Licence
Social	AlphaBay full-site archive	114,231 active listings; 1,217,388 feedback records	Gwern-curated; no explicit licence
Social	Agora market archive	109,689 listings	Gwern-curated; no explicit licence
Social	Kilos cross-market review panels	235,326 reviews	community mirror
Social	Drug-listing corpus	194,433 listings	Gwern-curated; no explicit licence
Financial	Elliptic++ transactions	203,769 nodes / 234,355 edges / 183 features	public dataset, research use
Financial	Elliptic++ addresses	822,942 addresses / 2,868,964 address–address edges / 56 features	public dataset, research use
Infrastructure	Abuse.ch URLhaus	58,310 takedown records	free for research, attribution requested
Infrastructure	Abuse.ch ThreatFox	105,999 IOC records	free for research, attribution requested
External context	Monthly global breach-posture reports	9 issues (2025-01 … 2025-11, 05 and 09 absent)	context only; not used as evidence

Graph learning supplies labelled illicit-transaction benchmarks and node-classification baselines — the Elliptic transaction graph and its GCN baseline [7], the address-level Elliptic++ with heterogeneous-GNN baselines [8], and purpose-built anti-money-laundering datasets [9, 10]. Our contribution is ecosystem measurement rather than classification, and our negative result shows that even that transfer is unsafe without instrument checks.

Network robustness theory explains why targeted removal beats random removal on synthetic or reconstructed graphs [11–14], while evaluation-methodology work documents inflated baselines and leaky splits [15–17]. *Negative control* and *pre-registration* are the classical names for the discipline we adopt [18, 19], and simulations of criminal organisations under intervention remain rule-based rather than data-driven [20].

C1: a validation-first protocol. Every candidate metric passes a validity screen, every instrument is paired with controls required to be capable of rejecting it, and every reported number carries the audit trail that lets a reviewer reproduce the control applied to it. **C2: a controlled measurement corpus.** Seven analyses span the social, financial and infrastructure layers. Three claims survived their controls and four did not: a headline "3.7× more persistent" claim was killed by an immortal-time-bias control (**-0.49 pp**), and a "+3.64 reviews/week" displacement estimate was demoted by its own non-centred placebo null to two non-significant point estimates. The magnitude is therefore bounded rather than estimated. **C3: a negative-result ledger.** Temporal-graph prediction of recurrence fails four independent instrument checks, traced to a feature-side separability trap that inflates shuffled-label controls, which de-leakage only partially repairs; the hazard, its diagnosis and its partial repair are kept as a result in their own right. The paper is organised as Data (§2), Protocol (§3), Controlled analyses (§4), Negative results (§5), and Governance, limitations and conclusion (§6–§7).

2 Data

The three layers share no time axis, and none of the datasets was collected for the purpose of measuring resilience, so every cross-layer comparison in this paper is an ordering rather than a ratio. All sources are public (Table 1): the social layer is a full AlphaBay archive with Agora, Kilos and drug-listing dumps, and the financial layer is the Elliptic++ Bitcoin graph whose 49 ordinal steps anchor from outside the data to approximately 2016-01-01 to 2017-11-03. The infrastructure layer is the Abuse.ch URLhaus and ThreatFox feeds, which are rolling-retention and current through 2026-09-23. Eight sources are public archives used under research terms, two are Gwern-curated archives with no explicit licence used under research fair use, and one context source is not used as evidence; no raw leaked personal records are touched and identifiers are hashed.

3 A Validation-First Measurement Protocol (C1)

No quantity enters this paper until it has passed three gates, in order, and the gates are stated in general terms rather than tailored to the one domain we demonstrate them on (Fig. 1).

G1: metric validity. A resilience metric must (a) recover a ground-truth planted value in synthetic data, (b) distinguish real measurements from randomised controls, and (c) respond to the construct it claims to measure. Six candidate resilience metrics were screened against three pre-registered criteria — rank recovery against a planted truth *without* collapsing onto another metric's truth, calibration whose slope interval covers 1 and intercept interval covers 0, and separation from a week-permuted null. **Each criterion passes two of the six metrics, and no metric passes all three.** The metric with the highest own-truth correlation (`recovery_time`, Spearman 0.9859) fails the rank criterion precisely because it is indistinguishable from `centrality_redistribution` (max |cross-Spearman| = 0.8407 $\geq$ 0.8). The two metrics that pass the rank criterion fail the permutation criterion by construction, since a time-invariant structural quantity cannot separate from a week-permuted control.

Reporting a failure is itself a result. Because the artifact lists the three criteria without stating a conjunction rule, any single count of survivors is a post-hoc choice, so we report the per-criterion matrix rather than a score (Supplementary Note S2). A metric that fails is either not reported or reported together with its failure; the six metric *names* come from the pre-registered plan while their operational definitions are ours, so none of this validates the delivered measure.

G2: instrument validity. An estimator or predictor must be paired with controls that *can* reject it: a negative-control band with a pre-registered usable window [18], permutation tests, and a shuffled-label control repaired against feature-side separability. Every design is registered before it is run [19]. A control whose value is fixed by construction is not a control, because G2 requires demonstrating that the control varies within the stratification the metric uses. We found that failure three times: in two single-feature arms whose retained feature is constant within every bin, and in a de-leakage configuration whose retained feature set is the stratifying variable itself. A control in that state fails G2 by definition rather than by a margin, and a design that reports it as a pass has not tested anything.

G3: auditability. Every reported number maps to a hash-verified artifact in a manifest, or is flagged as not yet in it, and cross-artifact quantities must be reproduced independently by two pipelines before they are used. The manifest's domain is the result artifacts this study produced and its time point is its own build; the boundary rule is reproduced with the manifest so that the count can be recomputed rather than taken on trust (Supplementary Note S1). Two pipelines independently reproduced an address count of **61,487**, the rare cross-artifact agreement that licenses a reading.

CPU and GPU replicates are never tabulated together: on one replicate the largest per-cell absolute difference in reported AUC was max $|d| = 0.0068$ against a pre-registered tolerance of 1e-3. Device-equivalence checks under fp64 deterministic mode reproduce bitwise across devices, whereas fp32 forward passes exhibit chaos amplification and are excluded from equivalence gates. Discipline of this kind is what makes a negative result interpretable, and what allows a failed instrument to be reported as a diagnosis rather than as a gap.

The two-list rule. When an instrument self-reports failure, the failure report itself is that gate's output and must be given: how many cells fell outside the window, which cells they were, what the window was, and the verbatim decision rule. What is forbidden is any quantity *read off* the failed instrument — a point estimate, a band, a comparison, a significance statement. The test for which list a reading belongs to is whether it can stand without the failed instrument: cell identities and the window cannot and are therefore reportable, whereas a shuffled-control AUC can and is therefore withheld.

The feature-side separability trap. G2 is where existing practice is thinnest, because a control that cannot reject anything looks exactly like a control that rejected nothing. In censored ecological graphs, labels are frequently derived from the same structural statistics that are also features — degree, transaction counts, address statistics — so shuffling the labels does not break the feature–label association and a shuffled-label control reads far above chance. The instrument that did not fail here is the permutation null: it uses **4000 draws**, one band per pre-registered horizon, and permutes time steps rather than nodes. On our data the shuffled-label control reads **0.71364** at $\Delta = 1$ against a permutation band of **[0.4852, 0.5148]**, and the instrument's verdict is OUTSIDE.

What anchors 0.5. A control whose score is independent of the label *by construction*, read through the same within-entry-step weighted estimator the arms are judged on, reads 0.4897–0.5141 (largest absolute deviation 0.01406). So on this data **0.5** is the right anchor and the shuffled control's distance from it is real. What that distance does *not* license is calling it leakage, because the excess conflates feature-side separability with label leakage, is not decomposable, and is therefore not published as any number.

How far the repair goes. Readings are within-entry-step weighted at $\Delta = 1$. Dropping the calendar column reads 0.71556 and dropping the position column reads 0.71758 — both *higher* than the as-built **0.71364**, so removing either column repairs nothing. The attribution is carried by the two arms that move the reading by more: the six family-mean features alone read 0.69372 and the two family-count features alone read 0.56115.

Retraining on reduced feature families does not return the control to chance: the strongest non-vacuous

configuration moves the reading from **0.71364** to **0.6561** and stops there (configuration-by-configuration accounting in Supplementary Note S3). Two arms read exactly 0.5, but that is a construction artefact: a single feature constant within every bin yields an AUC of exactly one half, so those are vacuous controls rather than evidence of cleanliness. The transferable lesson is that in this measurement domain a shuffled-label control is not automatically a null model, and the first thing to check is whether it can vary at all.

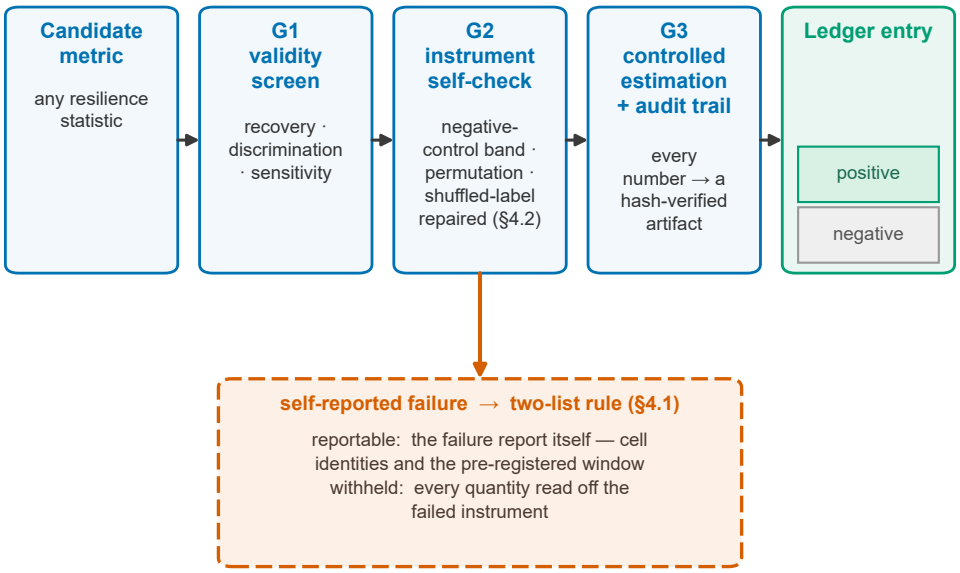


Figure 1: The five-step protocol. A candidate metric enters the G1 validity screen and, if it survives, the G2 instrument self-check, then controlled estimation with its audit trail (G3), ending in a ledger entry that records a positive or a negative result. The red side branch is the protocol's most distinctive move: when an instrument self-reports failure at G2, the failure report itself becomes that gate's output under the two-list rule, while every quantity read off the failed instrument is withheld.

4 Controlled Analyses (C2)

Each of the seven analyses follows the same four-part structure — the question, the design, the control that bites, and the finding with its bound — and in four of the seven the control that bites took down a claim of our own. One caveat belongs before the first of them: the seven are not seven independent studies. The displacement analysis in §4.4 and the exploratory cross-layer association (Supplementary Note S2) share a panel, an estimator and the same November 2019 event, so the latter is in substance an audit of the former that found the pre-trend the earlier design had missed. The number of independent events behind this corpus is therefore smaller than seven. Table 2 gives the whole corpus at a glance; full controls and every rejected estimator are in Supplementary Note S2.

4.1 Host reconstitution after infrastructure takedown (E7-inf)

Question. After an IOC is taken down, how quickly does the hosting infrastructure behind it reappear?

Design. Abuse.ch takedown records; hosts and domains tracked across takedown events; reconstitution latency measured with stratified matched floors for chance re-appearance and within-host pairing for infrastructure reuse.

The control that bites. A naive comparison suggested that reusing infrastructure confers 3.7× more persistence. A within-host paired design, which removes the immortal-time bias that creates the illusion, reverses the sign to **-0.49 percentage points** (sign test p = 8.1e-07), and the small negative effect strengthens as bulk hosts are dropped. The naive number is a bias exhibit, not a finding.

Finding. Median takedown delay spans **0.47–1.52 days** across treatments that bracket the unobserved times. Fully darkened hosts do come back: **13.88%** return within one day, **6.2**× the stratified matched chance floor of **2.24%**, and about 21% within a week. Domain-hosted infrastructure returns more often than bare-IP infrastructure at equal host size (**80.77% vs 61.49%** at more than 20 URLs, a factor of 1.31, against a paired

Table 2: The seven controlled analyses: the quantity that survived each one, the layer it belongs to, the control that bit, and what that control rejected. Four of the seven controls took down a claim of our own.

Analysis	Layer	Headline quantity	The control that bites	What it rejected
E7-inf Host reconstitution after takedown	Infrastructure	**13.88%** of hosts return within 1 d vs a **2.24%** matched chance floor (**6.2**×)	Within-host paired design (removes immortal-time bias)	The naive **3.7**× advantage; controlled, **-0.49 pp** with the sign reversed (p = 8.1e-07)
E0 Survival after listing	Infrastructure	Median online lifetime **1.36 days**; S(1 d) = **0.6064**, S(7 d) = **0.1575**, S(30 d) = **0.0497**	Site- and reporter-clustered SEs	Concentration proposition: **p = 0.1736** by reporter, **p = 0.5978** by block, both non-significant; IID understated the SE by **14.44**×
E4 Product survival in the social layer	Social	Median **222.9 days** overall vs **10.8 days** reviewed; the gap *is* the censoring	Proportional-hazards test; outcome definition	The model read as a transaction process: the outcome is first recorded feedback, and proportional hazards is violated for **19 of 24** covariates
E1-v3 Directional displacement after a market seizure	Social	Exposed vendors appear on Apollon **+10.5 pp** above matched controls; the Empire cell is **+0.0 pp**, a zero denominator	Non-centred placebo null (500 draws)	The intensive-margin estimate **"+3.64 reviews/week"**; its own null is centred at **+1.1156**, a parallel-trends violation, so it is demoted; corrected magnitude **+1.0 to +2.4**
E5-fin Financial-layer recurrence	Financial	Recurrence **7.4716%** (n = 61,487), graded across label classes	Specificity test across two enforcement-indistinct classes	The causal reading of the gradient: the licit–unknown gap (**-2.470 pp**) nearly equals the illicit–licit gap (**-2.735 pp**)
E6-fin v2 Structural position of the illicit subgraph	Financial	Random removal must delete **60.0%** of nodes to halve the giant component; degree-targeted removal does it at **2.5%**	Saturation check; pre-specified window test	The dismantling reading: saturated at **47 of 49** steps (sd = 0.000043), so the metric cannot register a shock; the pre-specified July 2017 window does not reject (**p = 0.0954**, **0.3458**)
E3 Causal attribution of resilience	Social ↔ Financial	No matched design reaches cluster-robust p < 0.05; the licensable result is an absence (H4: β **= -0.5804, p = 0.8047**)	Propensity-score overlap; direct measurement of the H2 substrate	The PSM design (**overlap = 0.000**, zero pairs, max $\lvert b \rvert$ = **20.61**) and the H2 bridging mechanism: **124** valid BTC addresses, **three** in Elliptic, all non-transactional

control significant at p < 0.001). One control here is vacuous by construction, and we report it as that rather than as a pass: hosts carrying a single URL *cannot* produce a later addition, and both kinds read exactly 0.00% against approximately 8% floors (Fig. 2).

Bound. These are reconstitution dynamics of observable infrastructure; covert migration off these hosts is not measurable from takedown records, and the feed carries an unknown admixture of scanner traffic. The result also survives a change of aggregation scale: the week-7 reconstitution rate is not an artefact of one window. Recomputed month by month across the **4** months that clear the 100-record floor, the spread between the highest and lowest monthly value is **7.667 percentage points**, against a 10,000-draw permutation null spanning **[0.3612, 2.9833]** (p < 1e-4).

4.2 Infrastructure survival after listing (E0)

Question. How long does malicious infrastructure survive once listed?

Design. 36,262 queued URLhaus records inside the retention window; event rate 93.2%; Kaplan–Meier survival after first appearance.

The control that bites. A headline hypothesis, that concentration (block-level hosting density) predicts longer survival, fails under clustered standard errors (**p = 0.1736** clustered by reporter, **p = 0.5978** clustered by infrastructure block) after the IID standard error turned out to understate the block-clustered one by **14.44**×. We report the null, not the story. A second apparent effect, significant survival differences across groups, reduces to a label-semantics difference between untagged placeholder and source-probe groups, and is disclosed as such.

Finding. Median online lifetime is **1.36 days**; S(1 d) = 0.6064, S(7 d) = 0.1575, S(30 d) = 0.0497, so half of listed URLs are gone within a day and a half while about 5% survive beyond a month. At the malware-family level the pattern is robust: Mozi-tagged URLs are taken down about 1.6× as fast as the rest (median 1.13 vs 1.81 days). Every duration here is a lower bound, because `last_online` records the last sighting of a URL rather than the instant it was taken down. Short median lifetimes therefore coexist with the fast reconstitution of §4.1: churn, not death, is the dominant infrastructure dynamic. The curve is also stable over the observation window: at the 14-day horizon the fitted logistic trend across cohort weeks is an odds ratio of 1.0016 per day (interval [0.9999, 1.0034], p = 0.070), that is, no trend.

4.3 Product survival in the social layer (E4)

Question. How long do listings survive on a major market?

Design. AlphaBay archive, n = 114,231 listings, event rate 45.9%; Kaplan–Meier and Cox survival with covariates.

The controls that bite. The outcome is first recorded feedback, not first transaction, so a listing that transacts without leaving feedback is censored rather than counted, and the 45.9% event rate is a lower bound on the transaction rate. The two medians must therefore be reported together, **222.9 days** overall versus **10.8 days** for the reviewed subset, because the gap *is* the censoring and quoting either alone misstates the result. Proportional hazards is violated for **19 of 24** covariates and no seller key exists, so we report Kaplan–Meier medians as descriptive and treat every Cox coefficient as exploratory.

Finding. Listing-level persistence in the social layer is long relative to infrastructure: median **222.9 days** against **1.36 days**, a ratio of about 164. The two layers are not on a shared clock, so this is an ordering of two different quantities, not a like-for-like duration ratio, and we use it only to say which layer is the slower one.

Bound. No seller-level clustering is possible, so these are descriptive quantities rather than effects, and the two social-layer panels that carry no date column (Kilos, drug-listing) cannot contribute to the comparison at all.

4.4 Directional displacement after a market seizure (E1-v3)

Question. After the November 2019 shutdown of Cryptonia and Cannazon, where did exposed vendors actually go?

Design. Exposure-matched comparison of vendor appearance rates on candidate successor markets; the final of three generations of this analysis, the two earlier designs having been declined by their own falsification tests.

The controls that bite. Two, and both cut. The aggregate picture (Apollon rising from 6.2% to 68.4% of ecosystem volume while Cryptonia and Cannazon go to zero) is exactly what a collector re-weighting its coverage would produce with nobody moving, so we report it as a description and never as an effect.

And the original intensive-margin estimate of "**+3.64 reviews/week**" was killed by its own placebo machinery: the 500-draw placebo null was itself centred at **+1.1156** (sd 0.3205), which *is* a parallel-trends violation, and the three placebos originally reported were all drawn below that null's mean. The largest of three draws is therefore a selection minimum rather than a characterisation of the null. The corrected magnitude, on the designs whose nulls sit near zero, is **+1.0 to +2.4 reviews/week** against a treated pre-period base of 1.15. One cell in the per-source breakdown has a zero denominator (**Empire: 0.0% against 0.0%**, odds ratio undefined) and is unmeasurable rather than a "no effect"; a second (**Cannazon**) rests on 10 events and is underpowered rather than confident.

Specification choice. We ask how much of the demoted estimate survives the choices a reasonable analyst would make. Twelve core specifications — varying the fixed-effect structure, the binning, the event definition, the target market and the pre/post window — are all estimable, and every one is positive and clears zero at the 5% level. The curve is anchored to the incumbent analysis rather than standing free, reproducing its coefficient exactly when the candidate set is held equal. The coefficients themselves are listed in Supplementary Note S2. What the curve is *not* is a rehabilitation of the magnitude: it varies the specification, not the placebo machinery, so it does not re-test the parallel-trends violation that demoted the estimate, and its breadth must not be read as an interval estimate.

Finding. Displacement is real and directional. Exposed vendors appear on Apollon **+10.5 percentage points** above matched controls, and on the shrinking market Empire **+0.0 points**, a zero-denominator cell that is unmeasurable rather than a measured null. From the Cryptonia-exposed population the odds ratio is **3.34 (p = 3.5e-6)**, against 0.968 for Cannazon-exposed. Two markets died on the same day; one population substituted, the other did not move.

Bound. Two bounds, neither decorative. The corrected figures are **two non-significant point estimates** (+1.0424 at cluster-robust p = 0.36 and +2.4426 at p = 0.24), so we report **no interval estimate**. What the placebo draws support is the existence of an effect, since the real coefficient sits in the 96.8th–100th percentile of its own null; what they forbid is any statement that the effect lies in a narrow band. And because treatment is defined as activity on a target market *and* on a market that died, part of the +10.5 points is the same sellers' activity re-registering rather than market-level substitution. So **+10.5 pp is an upper bound on substitution**; the qualitative point, that displacement has addresses and not merely volume, is the robust content.

4.5 Financial-layer recurrence (E5-fin)

Question. After a node is flagged illicit in Elliptic++, how often does activity recur through it?

Design. Recurrence of participation over the 49-step ordinal axis, by label class, estimated within entry step so that the differing amount of room available to late entrants is removed exactly rather than modelled.

The control that bites. The headline gradient (illicit **1.816%** < licit **5.796%** < unknown **8.371%**; **7.4716%** overall) is not licensable as a class effect. A specificity check comparing two classes that enforcement does not distinguish shows the licit–unknown gap (**-2.470 pp**) nearly equals the illicit–licit gap (**-2.735 pp**). A monotone gradient across all three labelling categories is therefore as consistent with how addresses entered the labelled subgraph as with anything about illicit behaviour. We report the gradient, withhold the causal reading, and withhold every p-value, because between-step differences span −27.4 pp to +13.1 pp, roughly 100× the nominal standard error, so the fixed-effect standard error assumes a homogeneity the data plainly violates.

Finding. Recurrence exists at **7.4716%** overall and is strongly graded. This quantity anchors here because two independent pipelines reproduced the classifying set at **61,487** identically, the rare cross-artifact agreement that G3 demands.

Bound. The labels are dataset-internal and cannot be certified against external ground truth, and the outcome is recurrence in the *labelled* subgraph rather than in Bitcoin, so an address "dies" here when it stops touching that subgraph.

4.6 Structural position of the illicit subgraph (E6-fin v2)

Question. What would network-level disruption of the financial layer actually require?

Design. Dismantling analysis on the Elliptic++ address graph as one aggregate snapshot (822,942 nodes), not on the 49 per-step snapshots: random versus degree-targeted node removal, and full removal of the illicit-labelled set. Two pre-specified tests accompany it.

The controls that bite. The giant-component metric itself nearly fails: at **47 of 49** time steps connectivity is saturated at 1.0000 (sd = 0.000043), so per-step giant-component size cannot register a shock at this resolution and the metric is usable only in the rare unsaturated steps. Separately, a pre-specified test of the July 2017 enforcement wave does not reject on any observable. The window was fixed at steps 41–43 before the run, the direction as "lower", and the null drawn from 20,000 random 3-step windows; the p-values are **0.0954** for illicit share and **0.3458** for illicit address count, with a third observable moving the *wrong* way. We report this as a low-power test, not as evidence of absence.

Finding. Random removal must delete about **60%** of nodes to halve the giant component; degree-targeted removal does it at **2.5%** (Fig. 3). The asymmetry is between the two *removal fractions* (60/2.5 ≈ 24), not between two effect sizes, and it carries the scale-free signature. Deleting **all** illicit-labelled nodes (**14,266** nodes, **1.734%** of addresses) leaves the giant component at **98.88%**, indistinguishable from removing the same number at random, and targeting the illicit subgraph's best-connected members does no better. The mechanism is visible in the degree table rather than inferred: mean degree **3.645** for the illicit class against **10.803** for the *licit* class, so the contrast is illicit against licit and not illicit against the whole graph.

Finding, continued: where the illicit class sits. The removal result is one reading of the illicit subgraph's position, and the nested-core profile agrees with it. The aggregate graph has degeneracy **768**, and the illicit class sits nearer the outside of that nesting than the licit class: mean coreness is 2.2114 for illicit addresses against 6.0496 for licit ones, a difference of -3.8382 against a 10,000-draw permutation null spanning [-0.1825, 0.1912] ($p < 1e\text{-}4$). The label is at the same time strongly assortative — label assortativity 0.7606 against a null spanning [-0.0027, 0.0019] — so the two readings are not in tension: the illicit label is very nearly a community of its own, and that community is peripheral in the nested-core sense rather than central.

Ranking four removal strategies by the area under the curve of cleared illicit share against node budget gives **34.2723** for random, **19.9398** for k-core-targeted, **5.2254** for degree-targeted and **1.4031** for PageRank-targeted, every comparison against random significant at the permutation floor over 16 seed pairs ($p < 1e\text{-}4$). Budget spent by coreness is therefore spent on the licit core rather than on the illicit periphery. The marginal-cost field, the node budget needed to clear 1% of the illicit set, reads 8229.4 for random, degree and PageRank alike and 65835.4 for k-core. That field is right-censored by its own grid — it takes the first grid point at which the threshold is met — so the ordering of strategies is carried by the AUC readings, not by this field.

Bound. Address-level enforcement against illicit finance has no network-level effect *in this graph*, and this is the labelled subgraph, whose composition is a collection artefact.

4.7 Causal attribution of resilience (E3)

Question. Which structural features cause faster recovery?

Design. Pre-registered hypotheses H1–H4 (decentralisation, cross-layer bridging, functional redundancy, centrality), tested by propensity-score matching plus difference-in-differences around the November 2019 shutdown, with 500 placebo event dates per design.

The control that bites. This analysis mostly did not survive. The pre-registered PSM covariate set is infeasible because it contains `n_markets`, which treatment *defines*, so treatment becomes perfectly predictable and matching returns **overlap = 0.000** with zero pairs (max $|b| = 20.61$) — a specification failure, not a vanishing effect.

Three reduced specifications match (121, 77 and 74 pairs) with point estimates **+1.04, +1.42 and +2.44** reviews/week, and the second column of each design gives +1.34, +3.64 and +4.49, for a range of +1.04 to +4.49. But the two largest second-column values stand on placebo nulls whose means are not centred at zero (**+0.6672** and **+0.6606** against a treated pre-period base of 1.15), which is the condition §4.4 uses to demote an estimate. On the column whose three nulls sit near zero, the usable range is **+1.04 to +2.44**. No matched design reaches cluster-robust $p < 0.05$, and the two uncertainty measures disagree on the same design (cluster-robust $p = 0.36$, placebo rank $p = 0.022$), so we report the divergence rather than choosing the smaller number.

H3 has no class column and is unmeasurable, and the substrate for H2 was measured and found not to exist: only **124** checksum-valid BTC addresses exist in the entire social layer, **three** intersect Elliptic, and all three are adjudicated non-transactional.

Finding. Of four hypothesised mechanisms, one returns a licensable result, and what it licenses is an absence rather than a positive relation: H4's centrality term is null (β **= -0.5804, p = 0.8047**), while H1's decentralisation term is positive but likewise non-significant (**+2.4823, p = 0.2416**). The asymmetry is the point: H1 predicted a positive interaction and H4 predicted no such relation, so the same non-significance fails one and supports the other. Matching being infeasible, the design falls back on 500 placebo draws per design, and every matched design places its real coefficient in the 96.8th–100th percentile of its own null, which supports "an effect exists" and forbids "the magnitude is such-and-such". We report the design's failure as a bound on what this data can ever say about mechanisms — with a single ecosystem-wide shock, effect magnitudes are not identified by any matching or placebo machinery.

4.8 Cross-analysis synthesis

Three findings survived their controls. (i) Churn dominates death at the infrastructure layer (1.36-day medians, 6.2× reconstitution) while the social layer persists (222.9-day medians), so resilience sits in the fast layer and persistence in the slow one. (ii) Displacement is directional: it has addresses, but the corrected figures are two non-significant point estimates (+1.04 and +2.44 reviews/week), not an interval. (iii) The illicit financial subgraph is a periphery, not a core, which bounds what address-level enforcement can achieve. These are not the same quantity measured twice: different populations, different censoring mechanisms and different event definitions, on layers that share no time axis.

5 Measurement Hazards and Negative Results (C3)

A measurement study is only as credible as its record of what failed, so we keep a ledger. The flagship entry is a fully diagnosed instrument failure rather than a missing result, and the diagnosis transfers beyond this dataset.

Temporal prediction of recurrence (E5p/E5r): four independent failures. Temporal-graph models — TGN [21] and the temporal-attention variant of the same family [22] — were asked to predict whether an **address in the labelled Bitcoin graph** recurs, that is, whether it is observed in at least two of the 49 ordinal time steps after its first appearance. They are set against MLP and GraphSAGE [23] baselines, with the instrument gates pre-registered. The target is a recurrence label on an ordinal axis, not the survival of a social-layer node and not anchored to an enforcement event, and every model sees only information available at the address's entry step.

1. **The predictor lost to its baselines.** On the adjudicated within-entry-step metric, all three TGN deltas fall below both baselines (-0.046 to -0.053 against MLP; -0.013 to -0.023 against GraphSAGE).
2. **The negative-control band self-reported failure, and its report is separated from its readings.** The band is its own instrument: 400 draws, nominal window [0.005, 0.055]. Its verbatim verdict is

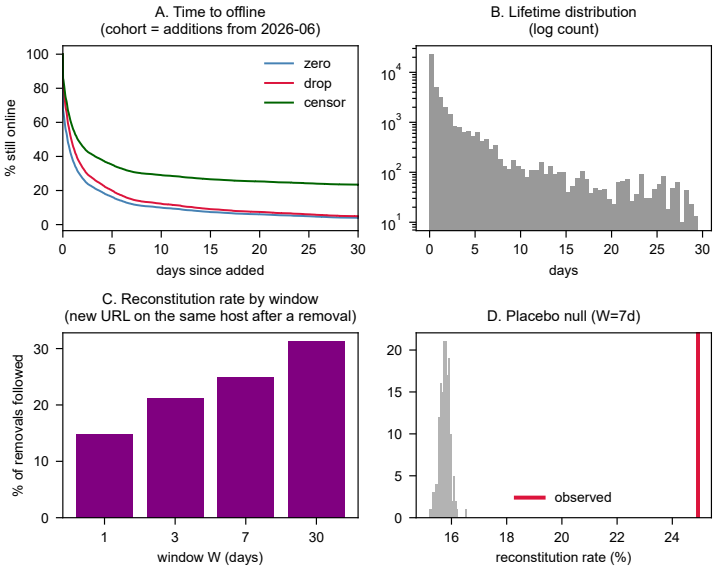


Figure 2: Reconstitution of hosting infrastructure after takedown (E7-inf), in four views. (a) Time from first sighting to going offline, for the cohort added from 2026-06. (b) The same lifetimes on a logarithmic count axis. (c) Reconstitution rate against the observation window. (d) Placebo null at $W = 7$ days, so the observed rate is read against its null and not against zero. Every duration is a lower bound.

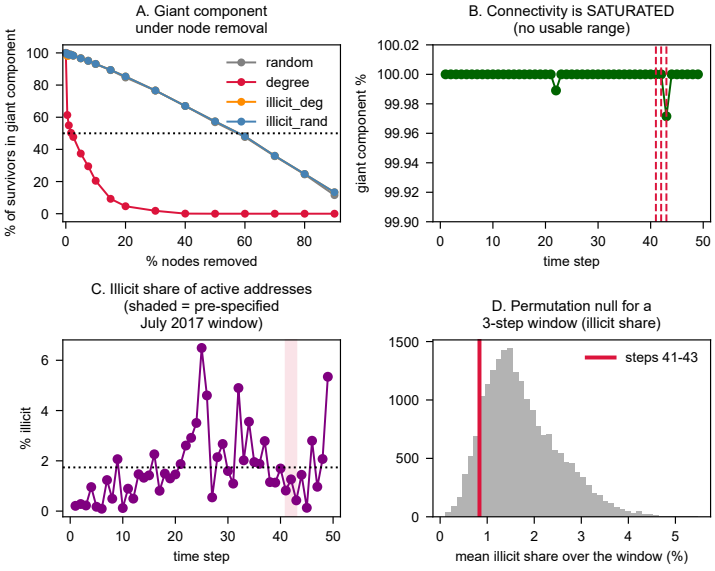


Figure 3: Dismantling the illicit subgraph (E6-fin), four views on the Elliptic++ address graph as one aggregate snapshot. (a) Giant component under node removal. (b) Connectivity across the 49 time steps, saturated and drawn to record that. (c) Illicit share of active addresses, with the pre-specified July 2017 window shaded (steps 41–43). (d) Permutation null for that three-step window.

INSTRUMENT FAILED, on exactly three cells (F1 G1 P1 global D=3 at rate 0.0700, F1 G1 P1 global D=5 at 0.0700, and F1 G1 P2 within-bin D=5 at 0.0650). Under the two-list rule that report is the gate's output and we give it in full. No quantity is read from *this* band, and in particular we publish no "leakage" estimate anywhere, because the excess of the shuffled control over chance conflates feature-side separability with label leakage and is not decomposable.

3. **The ablation gate failed, and the family attribution stands on a different instrument.** The pre-registered ablation gate did not pass, and a CPU–GPU replicate exceeded tolerance (max $|d| = 0.0068 >$ 1e-3), so the ablation *estimates* are never tabulated alongside the CPU-series numbers. That is a prohibition on quantities, not on diagnosis: the per-family attribution in §3 is read from the ablation *arm readings*, which carry their own decision rule, not from the failed estimate.
4. **The apparent positive result is not licensable.** The increment of the true model over the shuffled control is +0.05, two point estimates with no null distribution, and we claim no significance for it.

The trap, measured rather than described. The feature matrix of the pre-registered plan contains a column that *is* the $\Delta = 1$ label (`num_timesteps_appeared_in` ≥ 2). A baseline consuming that matrix reaches test AUC **1.0**, and the artifact declares the same baseline inadmissible in the same breath, for exactly that reason. We agree with the declaration and keep the number because it measures something else — how completely a structurally derived label can be re-derived from its own feature table. It is a measurement of the hazard, not a performance figure.

How far the repair goes. Read on the within-entry-step metric the arms are judged on, the three de-leakage configurations are not what a summary of them suggests. Configuration L1 keeps nine of the as-built arm's ten features (all but the calendar column) and reads **0.7155632**, *above* the as-built baseline of **0.7136417**, so dropping the calendar channel alone repairs nothing. Configuration L2 keeps three features and reads **0.6560979**, a real reduction that still does not reach chance. Configuration L3 reads exactly **0.5000** and the artifact marks it at chance — but L3 retains only the stratifying variable itself, and a single feature constant within every bin yields an AUC of exactly one half by construction. L3 is therefore a vacuous control in the sense §3 defines, and its 0.5 is not evidence of repair. **The strongest non-vacuous configuration moves the control from 0.7136 to 0.6561 and stops there.** (Fig. 4) The values 0.633 / 0.652 / 0.681 that the same artifact reports are its *pooled* per-horizon figures, which the artifact itself notes are not comparable to 0.5.

One limitation is declared inside the artifact and repeated here: the de-leakage removes node and tabular feature columns, but **graph message passing still runs over the full snapshot graph**.

A floor worth recording. An untrained model of the same architecture already separates at up to **0.6378** (GraphSAGE, $\Delta = 1$), and across the three configurations the untrained GraphSAGE readings span **0.3672**–**0.6378**, so any AUC inside that band is architecture rather than learning. No reported reading is treated as a performance figure until it clears that floor, which is the same discipline §3 applies to the metric layer and the reason the ledger does not stop at a bare failure count.

Why this is in the paper. Within the scope of this data structure and this label-construction process, an evaluation that shuffles labels without first demonstrating that the shuffle breaks the feature–label association *inside each stratification bin* can report an inflated AUC with a clean conscience and a broken null model. Whether that transfers to other censored, structurally labelled graphs is a question this paper does not answer. The feature-side separability trap is an instance of the leakage family that Kapoor and Narayanan [16] catalogue, and the de-leakage procedure and the vacuous-control test of §3 are the paper's reusable countermeasures.

A graph-side de-leakage stage, in which message passing was restricted to within entry-step strata, was voided by its own pre-registered replication gate: the mask-off control cells failed bitwise agreement with the prior generation, and by the registered rule all readings of that round are withdrawn. A directed attribution run reproduced the non-temporal models bitwise and isolated the discrepancy to the temporal model's environment-level nondeterminism, and an independent cross-device comparison reproduced that attribution under fp64 deterministic mode, where the non-temporal arms agree bitwise across CPU and GPU while the temporal arm does not. Cross-environment replication of temporal-graph pipelines is itself a hazard the protocol is designed to intercept, and this entry records it doing so.

Three further dead ends, briefly. Unsupervised changepoint detection on temporal embeddings of the market graphs (E14) missed every reference event, reading a firing rate of **0/25 = 0.000** and clearing its threshold by missing rather than by detecting, while a volume-only control does fire (rate 0.040), so the detector is not inert. The originally specified events lie outside the data's time range, and the substitutions actually used are disclosed in Supplementary Note S3. Two-event consistency (E2) is infeasible by pre-registered criterion, because no acquired dataset brackets Operation Onymous (2014-11) or Bayonet (2017-07), which is why §4.4 measures the November 2019 shutdowns instead. Role annotation (E10) never reached data acquisition and

no number here depends on it, and external validation against monthly breach-posture reports (EV-legacy) was rejected as a validation target for window drift.

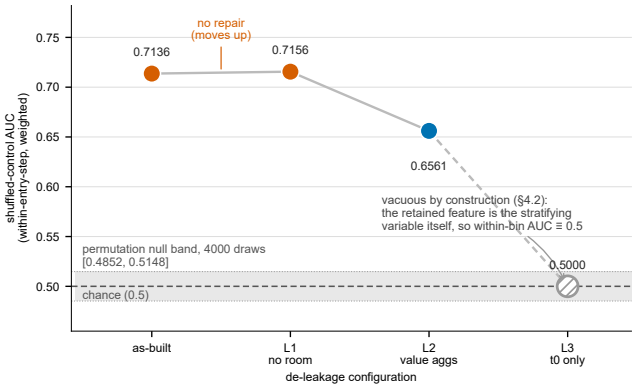


Figure 4: The shuffled-control trajectory, trap versus post-de-leakage, on the within-entry-step weighted AUC. The as-built configuration reads 0.7136; dropping the calendar column alone reads 0.7156, *above* it, so the path moves the wrong way before it moves right. The strongest non-vacuous configuration reaches 0.6561 and stops. The final point reads exactly 0.5000 and is drawn hollow and hatched because it is fixed by construction, not achieved.

6 Governance Implications, Conditional on the Bounds

Every implication below is conditional on the bounds of §4; none survives transposition into an unconditional policy claim.

1. **Targeted dismantling is the only lever that moves the structure.** Random removal needs about 60% of the financial graph to halve its giant component whereas degree-targeted removal needs 2.5% (E6-fin). *Condition:* the giant component is a weak proxy at 47/49 saturated steps, so this is about relative leverage, not absolute effect size.
2. **The illicit financial subgraph is a periphery, so act accordingly.** Removing 100% of illicit-labelled addresses leaves the component at 98.88% (E6-fin). *Condition:* address-level enforcement is disruption of service rather than damage to the network, and expecting network-level effects from it is a category error our data bounds.
3. **The fast layer is where resilience lives.** 13.88% of darkened hosts return within a day at 6.2× chance (E7-inf) and median infrastructure lifetime is 1.36 days (E0). *Condition:* takedown cadence, not takedown existence, is the operative variable at this layer.
4. **Domain infrastructure is the durable substrate.** Domain-hosted infrastructure returns 1.31× as often as bare IP (80.77% vs 61.49%, $p < 0.001$). *Condition:* interventions that raise the cost of domain re-registration attack the slower half of the churn cycle, and the feed's scanner admixture is not separable.
5. **Displacement is directional: single-market seizures relocate rather than remove.** Exposed vendors surface on Apollon at +10.5 pp above controls and on Empire at +0.0, a zero-denominator cell that is unmeasurable rather than a measured zero (E1-v3). *Condition:* the corrected figures are two non-significant point estimates, so the directional design and not the magnitude is the finding, and +10.5 pp is an upper bound on substitution.

7 Limitations

We consolidate, rather than disperse, the weaknesses a reviewer would find.

(i) **Causal magnitudes are not identified.** E3's design failed at matching (overlap = 0.000), and with one ecosystem-wide shock no placebo or matching machinery recovers an effect size; only H4 survives, and H4's support is the confirmation of an absence, which is weaker evidence than a positive finding. This is not a null result about the mechanism; it is the demonstration that this data cannot support effect-size estimation in any design we could build.

(ii) **Historical windows.** The dated parts of the social and financial layers cover roughly 2014–2017: the AlphaBay listings span 2014-12-21 to 2017-01-28, Agora 2014–2015, and the financial layer's ordinal axis maps to approximately 2016-01-01 to 2017-11-03. Two social-layer panels (Kilos, drug-listing) carry no date column at all, so no window can be stated for them, and the current layer is infrastructure only.

(iii) **Censoring.** The social-layer outcome is first recorded feedback rather than first transaction, and no seller key exists, so the event rate is a lower bound and the standard errors are optimistic (E4). Both defects push every duration estimate in the same direction, which is why the two medians are always reported as a pair.

(vi) **Label provenance.** Elliptic++ labels are dataset-internal and unverifiable against external ground truth, and the E5-fin gradient is equally consistent with collection economics; the metrics that rest on this labelling layer inherit this base.

Dropped limitations (metric-validity detail, external validation, novelty exhaustiveness) are retained in Supplementary Notes S2 and S5.

8 Conclusion

We set out to measure the resilience of illicit data-trading ecosystems and ended up measuring, just as much, the reliability of the measurement itself. The protocol gates metrics before they are used, pairs every instrument with controls that can actually reject, hash-manifests every reported number, and keeps a public ledger of what failed. The controlled findings that survive are modest in number and better attested than usual: churn-dominant infrastructure with 6.2× reconstitution, a directional displacement whose magnitude is bounded rather than estimated, and a structurally peripheral illicit financial subgraph that can be removed in its entirety without moving the network's core. They also rest on a thin dimensional base, since no resilience metric cleared all three validity gates, and in a field where the resilience claims we build on rest on measurements no control has tested, we accept that trade.

Supplementary information

The Supplementary Materials contain the complete audit trail; every number in the main text can be traced to a note below. Five notes are supplied as a separate file (`supplementary.tex`).

- **Supplementary Note S1.** Complete protocol detail: the two-list rule, the audit census, and the engineering discipline.
- **Supplementary Note S2.** The seven analyses' full controls and rejected estimators, the specification-curve coefficients, the exploratory cross-layer association, and the metric-validity matrix.
- **Supplementary Note S3.** Per-configuration anatomy of the negative results (L1/L2/L3 de-leakage, the untrained floor, the changepoint and infeasibility entries).
- **Supplementary Note S4.** Family-level infrastructure profile, including the E0 survival panel (Supplementary Fig. S1), and the scenario-robustness analysis.
- **Supplementary Note S5.** Data, time coverage and licence supplement, including the external-context source and the ethics details.
- **Additional files 1 and 2.** The rule that draws the census boundary, and the SHA-256 census of the 302 result artifacts, as two plain-text files (`Additional file 1.txt`, `Additional file 2.txt`).

References

[1] K. Soska and N. Christin. Measuring the longitudinal evolution of the online anonymous marketplace ecosystem. In *24th USENIX Security Symposium*, pages 33–48, 2015.

[2] A. ElBahrawy, L. Alessandretti, L. Rusnac, D. Goldsmith, A. Teytelboym, and A. Baronchelli. Collective dynamics of dark web marketplaces. *Scientific Reports*, 10(1):18827, 2020. doi: 10.1038/s41598-020-74416-y.

[3] R. S. van Wegberg and T. Verburgh. Lost in the Dream? measuring the effects of Operation Bayonet on vendors migrating to Dream Market. In *WebSci'18*, pages 1–5, Amsterdam, 2018.

[4] D. Décary-Hétu and L. Giommoni. Do police crackdowns disrupt drug cryptomarkets? A longitudinal analysis of the effects of Operation Onymous. *Crime, Law and Social Change*, 67(1):55–75, 2017. doi: 10.1007/s10611-016-9644-4.

[5] P. Goonetilleke, A. Knorre, and A. Kuriksha. Hydra: lessons from the world's largest darknet market. *Criminology & Public Policy*, 22(4):735–777, 2023. doi: 10.1111/1745-9133.12647.

[6] L. Norbutas, S. Ruiter, and R. Corten. Reputation transferability across contexts: maintaining cooperation among anonymous cryptomarket actors when moving between markets. *International Journal of Drug Policy*, 76:102635, 2020. doi: 10.1016/j.drugpo.2019.102635.

[7] M. Weber, G. Domeniconi, J. Chen, D. K. I. Weidele, C. Bellei, T. Robinson, and C. E. Leiserson. Anti-money laundering in Bitcoin: experimenting with graph convolutional networks for financial forensics. In *Tutorial, Anomaly Detection in Finance Workshop, 25th SIGKDD*, 2019. arXiv:1908.02591.

[8] Y. Elmougy and L. Liu. Demystifying fraudulent transactions and illicit nodes in the Bitcoin network for financial forensics. In *KDD '23*, pages 3979–3990, 2023. doi: 10.1145/3580305.3599803.

[9] Y. Xiang, Y. Lei, D. Bao, et al. BABD: a Bitcoin address behavior dataset for pattern analysis. *IEEE Transactions on Information Forensics and Security*, 19:2171–2185, 2024. doi: 10.1109/TIFS.2023.3347894.

[10] H. Schnoering and M. Vazirgiannis. Bitcoin research with a transaction graph dataset. *Scientific Data*, 12(1):404, 2025. doi: 10.1038/s41597-025-04684-8.

[11] R. Albert, H. Jeong, and A.-L. Barabási. Error and attack tolerance of complex networks. *Nature*, 406 (6794):378–382, 2000. doi: 10.1038/35019019.

[12] X.-L. Ren, N. Gleinig, D. Helbing, and N. Antulov-Fantulin. Generalized network dismantling. *Proceedings of the National Academy of Sciences*, 116(14):6554–6559, 2019. doi: 10.1073/pnas.1806108116.

[13] S. W. Duxbury and D. L. Haynie. The responsiveness of criminal networks to intentional attacks: disrupting darknet drug trade. *PLoS ONE*, 15(9):e0238019, 2020. doi: 10.1371/journal.pone.0238019.

[14] L. Cavallaro, A. Ficara, P. De Meo, G. Fiumara, S. Catanese, O. Bagdasar, W. Song, and A. Liotta. Disrupting resilient criminal networks through data analysis: the case of Sicilian Mafia. *PLoS ONE*, 15 (8):e0236476, 2020. doi: 10.1371/journal.pone.0236476.

[15] O. Shchur, M. Mumme, A. Bojchevski, and S. Günnemann. Pitfalls of graph neural network evaluation. arXiv:1811.05868, 2018.

[16] S. Kapoor and A. Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804.

[17] F. Errica, M. Podda, D. Bacciu, and A. Micheli. A fair comparison of graph neural networks for graph classification. arXiv:1912.09893, 2019.

[18] M. Lipsitch, E. Tchetgen Tchetgen, and T. Cohen. Negative controls: a tool for detecting confounding and bias in observational studies. *Epidemiology*, 21(3):383–388, 2010. doi: 10.1097/EDE.0b013e3181d61eeb.

[19] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114.

[20] D. Manzi and F. Calderoni. An agent-based model for assessing the resilience of drug trafficking organizations to law enforcement interventions. *Journal of Artificial Societies and Social Simulation*, 27(3):3, 2024. doi: 10.18564/jasss.5430.

[21] E. Rossi, B. Chamberlain, F. Frasca, D. Eynard, F. Monti, and M. Bronstein. Temporal graph networks for deep learning on dynamic graphs. arXiv:2006.10637, 2020.

[22] D. Xu, C. Ruan, E. Körpeoğlu, S. Kumar, and K. Achan. Inductive representation learning on temporal graphs. arXiv:2002.07962, 2020.

[23] W. L. Hamilton, R. Ying, and J. Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems 30*, pages 1024–1034, 2017. arXiv:1706.02216.

Declarations

Availability of data and materials. All source datasets are public and are listed in Table 1. The analysis pipeline and the audited result artifacts are openly available at `https://github.com/chenyongjingg/illicit-data-economy-resilience` (tag `v1.0`). The rule that fixes the census boundary of the result manifest, and the SHA-256 manifest of all 302 result artifacts, are supplied with this submission as two plain-text supplementary files (Additional file 1 and Additional file 2) and are carried in that repository as well; Supplementary Note S1.3 summarises the census and Supplementary Note S5 records the licence detail.

Competing interests. The authors declare that they have no competing interests, financial or non-financial. No author holds any interest in, or has received any payment from, any organisation that would benefit from or be harmed by the results reported in this manuscript.

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Author contributions. Z.X., Y.C. and S.Z. designed the study. Z.X. and Y.C. assembled the data and ran the analyses. S.Z. supervised the work. All authors wrote and approved the manuscript.

Ethics statement. The study uses only public metadata and public archives; no leaked personal records were accessed; identifiers are hashed; no interaction with illicit actors occurred. Two Gwern-curated sources carry no explicit licence and are used under research fair use (Supplementary Note S5). LLM-based tooling was used in exploratory analysis only, and no reported number depends on LLM-generated labels. An early packaging artifact contained a default service credential; it was rotated and purged from all packaged snapshots.

Additional file 1. The rule that draws the census boundary (which trees are walked, what counts as an artifact, and the per-tree counts that sum to 302, so that the census count can be recomputed rather than taken on trust). *File format:* plain text. *Supplied as* `Additional file 1.txt`; summarised in Supplementary Note S1.3.

Additional file 2. The census itself: the SHA-256 digest of each of the 302 result artifacts, one per line. *File format:* plain text. *Supplied as* `Additional file 2.txt`.